AIORI HACKATHON 3

6G & FUTURE NETWORKS

# A3-PS018-TC100

## HAPS Link-Quality Prediction Using a Documented Channel Model

**Team FLUX**

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October 8, 2026

github.com/srinidhiishetty05/A3-PS018-TC100

# The Problem

- HAPS fly at about 20 km altitude and serve wide areas from a quasi-stationary position, a key non-terrestrial pillar of 6G.
- Received power and SNR swing by tens of dB with elevation angle, slant range, atmospheric gases and rain.
- A plain link budget misses this variability. A pure black-box model is opaque and fails outside its training conditions.
- Challenge: build a reproducible, model-based dataset for a fixed HAPS geometry, predict received power or SNR, and beat an analytical baseline on held-out trajectories.
- Scope: one band (28 GHz), one altitude (20 km), two scenarios. Synthetic and model-based only, with no real-network validation claim.

OUR ANSWER

**Physics generates the data. A hybrid ML model learns what the baseline misses, and reports how sure it is.**

# Background

## 01 Free-space loss

Grows with slant range: about 147 dB straight below the HAPS and about 156 dB at 50 km ground range, at 28 GHz [2].

## 02 Gas and cloud absorption

Oxygen, water vapour and clouds add a small, elevation-dependent loss [3][6].

## 03 Rain attenuation

The dominant impairment at Ka-band. It can add tens of dB in heavy rain and changes quickly in time [4][5].

## 04 Scintillation and pointing

Fast fluctuations of the received level on top of the slow trend [5].

**20 km**

HAPS altitude

**28 GHz**

Ka-band carrier

**20° to 90°**

elevation across the footprint

**Geometry alone moves the budget by about 9 dB. Rain can add tens of dB more.**

# Literature & Standards

[1]

### ITU-R P.528

Propagation curves for aeronautical services. The reference model named in the problem statement; used as a cross-check wherever its validity range applies.

[2]-[6]

### ITU-R P.525, P.676, P.838, P.618, P.840

The documented building blocks of our ground truth: free-space, gas, rain, scintillation and cloud attenuation.

[7]

### 3GPP TR 38.811

Shows how non-terrestrial channels are parameterised for NR.

[10][11]

### Hybrid residual learning

A physics prior plus a learned correction generalises better than a black box. We use gradient boosting [10] and ITU-Rpy for the ITU-R components [11].

[8][9]

### Quantile regression and conformal prediction

Calibrated intervals with finite-sample coverage under exchangeability. We test how they hold on unseen elevations.

THE GAP

**Most work uses either a purely analytical budget or a purely data-driven predictor. We combine both, validate on whole held-out trajectories, and attach calibrated uncertainty.**

# Proposed Solution

## 01

## GENERATE

Ground truth

- Layered ITU-R model: free-space loss, gases, rain, cloud, scintillation and pointing jitter.
- Seeded stochastic rain-rate series, so every run is reproducible.
- Two scenarios: clear sky and monsoon rain.
- Ground users move on many trajectories across the footprint.

## 02

## PREDICT

Baseline vs hybrid model

- Baseline: deterministic link budget with a fixed average atmospheric loss.
- Proposed: gradient-boosted residual model. SNR = baseline + learned correction.
- Inputs: elevation, slant range, scenario and noisy weather observations.

## 03

## TRUST

Uncertainty and validation

- Quantile regression plus split-conformal gives 90% prediction bands.
- Test on whole held-out trajectories, including an unseen low-elevation band.
- Report MAE against the baseline, interval coverage and a sensitivity ranking.

**LINK BUDGET (design assumptions)**

28 GHz carrier · 100 MHz bandwidth · 10 W (40 dBm) Tx · 30 dBi HAPS antenna · 35 dBi user antenna · 4 dB noise figure · -90 dBm noise floor

# Optimization Proposed

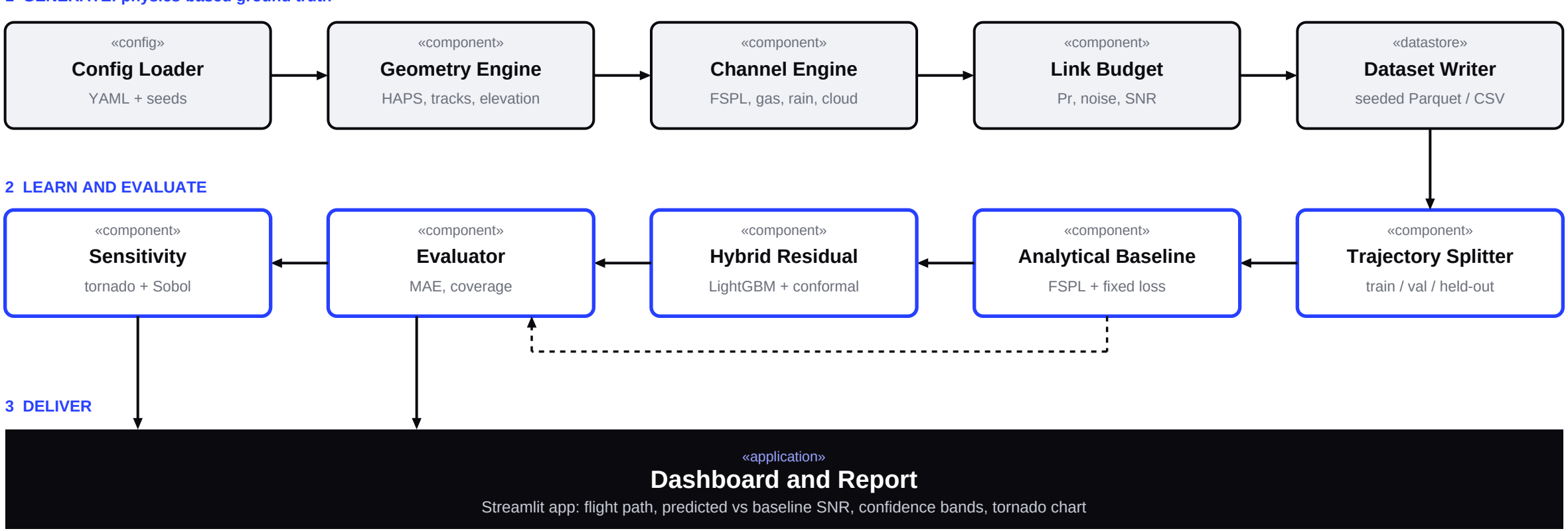
| ASPECT | BEFORE: analytical baseline | AFTER: proposed hybrid |
|---|---|---|
| **Atmosphere** | Fixed average loss for every condition | **Layered gas, cloud and rain model, with the residual learned from weather observations** |
| **Prediction** | Deterministic point estimate | **Hybrid residual prediction plus a calibrated 90% interval** |
| **Validation** | Random sample split | **Whole held-out trajectories and an unseen low-elevation band** |
| **Assumptions** | Implicit and undocumented | **Documented units and gains, with a tornado and Sobol sensitivity ranking** |
| **Reproducibility** | Ad hoc scripts | **Fixed seeds, YAML configs and a one-command rerun** |

**TARGETS, NOT RESULTS**

Lower MAE than the analytical baseline in both scenarios, with at least 90% empirical interval coverage on held-out trajectories. Measured numbers follow in Level 2.

# Solution Architecture

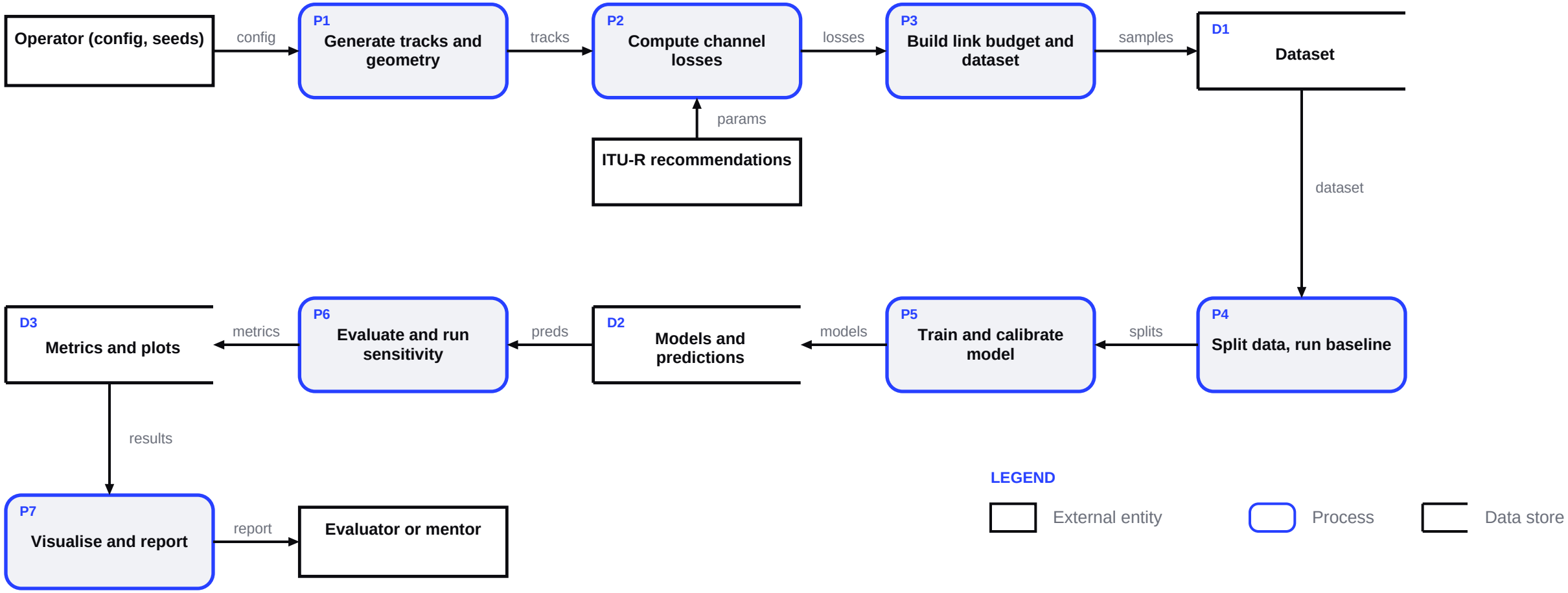
UML component view of the Python pipeline. Solid arrows are data flow; the dashed arrow carries baseline predictions.




Everything is configured from one YAML file and re-runs end to end from a fixed seed.

# Data Flow Diagram

Level-1 view. Every process reads and writes through a named store, so any result can be traced and re-run.



# Deployment Timeline

**8-14 Oct**

**LEVEL 1**

Repo, deck and intro video (Steps 1-7)

**16-20 Oct**

**MODEL**

Channel engine, link budget and baseline

**20-24 Oct**

**DATA**

Seeded generator and trajectory split

**24-28 Oct**

**LEARN**

Hybrid model, bands and sensitivity

**28-31 Oct**

**SHIP**

Dashboard, report and live demo (Steps 9-10)

**Nov-Dec**

**FINALS**

Evaluator connect, Regional 24-30 Nov, Finals 17 Dec

**HOW WE WORK**

Two-person team. Feature branches with pull-request review, a 15-minute daily sync, and mentor feedback at every milestone. Mentoring sessions run 16-31 Oct.

**ROLES**

**Srinidhi (Leader): RF and channel lead. Propagation models, link budget, baseline, sensitivity analysis, technical report.**

**Kaushik: data and ML lead. Dataset pipeline, hybrid model, evaluation, dashboard and live demo.**

# Bibliography

[1] ITU-R, Recommendation P.528, "Propagation curves for aeronautical mobile and radionavigation services using the VHF, UHF and SHF bands."

[2] ITU-R, Recommendation P.525, "Calculation of free-space attenuation."

[3] ITU-R, Recommendation P.676, "Attenuation by atmospheric gases and related effects."

[4] ITU-R, Recommendation P.838, "Specific attenuation model for rain for use in prediction methods."

[5] ITU-R, Recommendation P.618, "Propagation data and prediction methods required for the design of Earth-space telecommunication systems."

[6] ITU-R, Recommendation P.840, "Attenuation due to clouds and fog."

[7] 3GPP, TR 38.811, "Study on New Radio (NR) to support non-terrestrial networks."

[8] R. Koenker and G. Bassett, "Regression quantiles," Econometrica, vol. 46, no. 1, pp. 33-50, 1978.

[9] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," arXiv:2107.07511, 2021.

[10] G. Ke et al., "LightGBM: A highly efficient gradient boosting decision tree," Advances in Neural Information Processing Systems (NeurIPS), 2017.

[11] I. del Portillo, "ITU-Rpy: a Python implementation of the ITU-R P. Recommendations," open-source software.

# Thank You

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